

Lecture 11: The Derivative of $\ln x$

GOAL: Differentiate $\ln x$ and introduce the technique of Logarithmic Differentiation. Finish the proof of the Power Rule.

Summary:

$$\begin{aligned} [\ln x]' &= \frac{1}{x} & [\log_b x]' &= \frac{1}{x \ln b} \\ [\ln f(x)]' &= \frac{f'}{f} & x^p &= e^{p \ln x} \end{aligned}$$

Example 1 Find y' if $y = \frac{490t}{(t^2+7)^3}$

$$[490t]' = 490$$

$$[(t^2+7)^3]' = 3(t^2+7)^2 \cdot 2t = 6t(t^2+7)^2$$

$$y' = \frac{490(t^2+7)^3 - 490t(6t)(t^2+7)^2}{(t^2+7)^6}$$

Goal Develop a tool to deal with this issue.

Derivative of $\ln(x)$

Recall $[e^{f(x)}]' = e^{f(x)} \cdot f'(x)$. We can use this to find $\frac{d}{dx}(\ln x)$

$$f(x) = \ln x$$

$$e^{f(x)} = e^{\ln x}$$

$$e^{f(x)} = x$$

$$\frac{d}{dx}(e^{f(x)}) = \frac{d}{dx}(x)$$

$$e^{f(x)} \cdot f'(x) = 1$$

$$f'(x) = \frac{1}{e^{f(x)}} = \frac{1}{e^{\ln x}} = \frac{1}{x}$$

Theorem ① $\frac{d}{dx}(\ln x) = \frac{1}{x}$

② If f is a differentiable function:

$$\frac{d}{dx}[\ln(f(x))] = \frac{1}{f(x)} \cdot [f(x)]' = \frac{f'(x)}{f(x)}$$

Example 2 Find y' if $y = \ln(x^2 - 1)$

METHOD 1: $y' = [\ln(x^2 - 1)]' = \frac{1}{x^2 - 1} \cdot 2x = \frac{2x}{x^2 - 1}$

METHOD 2: $y = \ln(x^2 - 1) = \ln[(x-1)(x+1)]$

$$y = \ln(x-1) + \ln(x+1)$$

$$y' = \frac{1}{x-1}(1) + \frac{1}{x+1}(1)$$

$$= \left[\frac{1}{x-1} + \frac{1}{x+1} \right]$$

$$= \frac{x+1 + x-1}{(x-1)(x+1)} = \frac{2x}{x^2 - 1}$$

Example 3 Find y' if $y = \ln\left(\sqrt{\frac{x+1}{x^2+4}}\right)$

$$y = \sqrt{\ln\left(\left[\frac{x+1}{x^2+4}\right]^{\frac{1}{2}}\right)} = \frac{1}{2} \ln\left(\left[\frac{x+1}{x^2+4}\right]\right)$$

$$= \frac{1}{2} [\ln(x+1) - \ln(x^2+4)]$$

$$y = \frac{1}{2} \ln(x+1) - \frac{1}{2} \ln(x^2+4)$$

$$y' = \frac{1}{2} \cdot \frac{1}{x+1} \cdot 1 - \frac{1}{2} \cdot \frac{1}{x^2+4} \cdot 2x$$

$$y' = \left[\frac{1}{2x+2} - \frac{x}{x^2+4} \right]$$

Example 4 Find y' if $y = \log x = \log_{10} x$

$$y = \log_{10} x \quad \text{Change of Base Formula} \quad \frac{\ln x}{\ln 10}$$

$$y' = \frac{1}{\ln 10} [\ln x]' = \frac{1}{\ln 10} \cdot \frac{1}{x} = \frac{1}{x \ln 10}$$

In general, for a base $b > 0$,

$$\frac{d}{dx} (\log_b x) = \frac{1}{x \ln b}$$

Logarithmic Differentiation Recall if $y = f(x)$

$$[\ln y]' = [\ln(f(x))]' = \frac{1}{f(x)} \cdot f'(x) = \frac{y'}{y}$$

We can use this to our advantage.

Ex 5 Find y' if $y = \frac{490t}{(t^2+7)^3}$

$$\ln y = \ln \left(\frac{490t}{(t^2+7)^3} \right) = \ln(490t) - \ln[(t^2+7)^3]$$

$$= \ln 490 + \ln t - \ln[(t^2+7)^3]$$

$$\ln y = \ln 490 + \ln t - 3 \ln(t^2+7)$$

$$\frac{y'}{y} = \frac{1}{t} - 3 \cdot \frac{1}{t^2+7} \cdot 2t$$

$$\frac{y'}{y} = \frac{1}{t} - \frac{6t}{t^2+7}$$

$$y' = y \left[\frac{1}{t} - \frac{6t}{t^2+7} \right]$$

$$y' = \frac{490t}{(t^2+7)^3} \left[\frac{1}{t} - \frac{6t}{t^2+7} \right]$$

$y = \frac{490t}{(t^2+7)^3}$

Example 6 Find y' if $y = (2x+1)^5 (x^4-3)^6$

$$\ln y = \ln [(2x+1)^5 (x^4-3)^6]$$

$$\ln y = \ln[(2x+1)^5] + \ln[(x^4-3)^6]$$

$$\ln y = 5 \ln(2x+1) + 6 \ln(x^4-3)$$

$\ln(a^b) = b \ln a$

$$\frac{y'}{y} = 5 \cdot \frac{1}{2x+1} \cdot 2 + 6 \cdot \frac{1}{x^4-3} \cdot 4x^3$$

$$\frac{y'}{y} = \frac{10}{2x+1} + \frac{24x^3}{x^4-3}$$

$$y' = \underbrace{y} \left[\frac{10}{2x+1} + \frac{24x^3}{x^4-3} \right]$$

$$y' = (2x+1)^5 (x^4-3)^6 \left[\frac{10}{2x+1} + \frac{24x^3}{x^4-3} \right]$$

Example 7 Find y' if $y = x^x$

$$\ln y = \ln(x^x) = x \ln x$$

$$\ln y = x \ln x$$

$$\frac{y'}{y} = 1 \cdot \ln x + x \cdot \frac{1}{x}$$

$$\frac{y'}{y} = \ln x + 1$$

$$y' = y [\ln x + 1]$$

$$y' = x^x [\ln x + 1]$$

Example 8 If $y = (x^2 - 8x + 2)\sqrt{25 - x^2}$,
find y' .

$$\ln y = \ln \left[(x^2 - 8x + 2) \sqrt{25 - x^2} \right]$$

$$\ln y = \ln(x^2 - 8x + 2) + \ln \sqrt{25 - x^2}$$

$$\ln y = \ln(x^2 - 8x + 2) + \frac{1}{2} \ln(25 - x^2)$$

$$\frac{y'}{y} = \frac{1}{x^2 - 8x + 2} (2x - 8) + \frac{1}{2} \cdot \frac{1}{25 - x^2} (-2x)$$

$$\frac{y'}{y} = \frac{2x - 8}{x^2 - 8x + 2} - \frac{x}{25 - x^2}$$

$$y' = y \left[\frac{2x - 8}{x^2 - 8x + 2} - \frac{x}{25 - x^2} \right]$$

$$y' = (x^2 - 8x + 2)\sqrt{25 - x^2} \left[\frac{2x - 8}{x^2 - 8x + 2} - \frac{x}{25 - x^2} \right]$$

Example 9 Find y' if $y = \frac{(x-2)^5 \sqrt{x^2+1}}{(x+3)^7}$

$$\ln y = \ln \left[(x-2)^5 \sqrt{x^2+1} \right] - \ln [(x+3)^7]$$

$$\ln y = \ln(x+2)^5 + \ln(\sqrt{x^2+1}) - \ln[(x+3)^7]$$

$$\ln y = 5 \ln(x+2) + \frac{1}{2} \ln(x^2+1) - 7 \ln(x+3)$$

$$\frac{y'}{y} = 5 \cdot \frac{1}{x+2} + \frac{1}{2} \cdot \frac{1}{x^2+1} \cdot 2x - 7 \cdot \frac{1}{x+3}$$

$$y' = y \left[\frac{5}{x+2} + \frac{x}{x^2+1} - \frac{7}{x+3} \right]$$

$$y' = \frac{(x+2)^5 \sqrt{x^2+1}}{(x+3)^7} \left[\frac{5}{x+2} + \frac{x}{x^2+1} - \frac{7}{x+3} \right]$$

Derivative of x^p

If p is rational ✓

If p is irrational, then x^p is only defined for $x > 0$

$$[x^p]' = [e^{\ln(x^p)}]' = [e^{p \ln x}]'$$

$$= e^{p \ln x} \cdot p \cdot \frac{1}{x} = x^p \cdot p \cdot \frac{1}{x} = p x^{p-1}$$